GRADE 7 ACCELERATED · UNIT 5

Unit 5 Review

Multi-Step Proportional Relationships **ANSWER KEY**

Name
Date
Period

Show your work. Circle your final answer.

HOW TO USE THIS. This review covers every learning target from the unit — more than the assessment does. Work it in parts. If you miss two or more in a part, that part is where to spend your study time. The reference sheet has everything you need for each part.

PART A. Scale Drawings

Lesson 1 · MA.7.GR.1.5 · 5 questions

1. A 5 cm side of a figure is 8 cm in its scale copy. Find the scale factor.

k = 1.6

copy ÷ original

2. A 16 cm side of a figure is 4 cm in its scale copy. Find the scale factor.

k = 0.25

the copy is smaller, so k is under 1

3. A scale copy is made at k = 1.5. A side of the original is 6 cm. Find the matching side of the copy.

9 cm

4. A drawing has a 65° angle. Its scale copy is made at k = 2. What is the matching angle in the copy?

- ○ A. 67°
- ○ B. 130°
- ○ C. 32.5°
- ● **D. 65°**

D

5. A map is drawn at 1 cm : 30 km. Two towns are 4.5 cm apart on it. How far apart are they?

135 km

drawing × rate

PART B. Scale Factor: Perimeter and Area

Lessons 2–3 · MA.7.GR.1.5 · 6 questions

6. A figure with perimeter 16 cm is scaled by k = 2.5. Find the copy's perimeter.

40 cm

perimeter × k

7. A figure with area 9 cm^2 is scaled by k = 2.5. Find the copy's area.

56.25 cm^2

area × k^2 = 9 × 6.25

8. A 4 cm by 6 cm rectangle is scaled by k = ½.

a. Find the copy's perimeter.

10 cm

20 × ½

b. Find the copy's area.

6 cm^2

24 × ¼

9. A figure has area 10 cm^2 and its scale copy has area 160 cm^2. Find k.

k = 4

160 ÷ 10 = 16 = k^2

10. A figure's perimeter is 12 cm and its copy's perimeter is 36 cm. What is the area ratio, copy to original?

9

k = 3 from the perimeters; the area ratio is k^2

11. Sam says tripling every side of a rectangle triples its area. Explain why not, with a counterexample.

A 2 by 5 (area 10) tripled is 6 by 15 (area 90): nine times, because area is two lengths and each is tripled.

the reason and the numbers

PART C. Real-World Scale Drawings

Lesson 4 · MA.7.GR.1.5 · 4 questions

12. A plan is drawn at ½ in : 1 ft. A wall is 7.5 in long on it. Find the real length.

15 ft

1 in : 2 ft; 7.5 × 2

13. A plan is drawn at 1 in : 6 ft. A real wall is 21 ft long. Find its length on the plan.

3.5 in

real ÷ rate

14. A model boat is built at 1 : 25 and is 32 cm long. Find the real boat's length in metres.

8 m

32 × 25 = 800 cm

15. **SSDD.** Every part below is about one floor plan drawn at ¼ in : 1 ft, on which a room is 3 in by 4 in. 1 ft = 0.305 m. Same surface, four different questions: a length, an area with the rate used twice, the scale factor as one number, and a conversion — the four moves of the unit on one room, so that a student learns to read the question and not the picture.

a. Find the real length of the wall drawn 3 in long.

12 ft

1 in : 4 ft; 3 × 4

b. Find the room's real area.

192 ft²

12 by 16; the rate used twice

c. Write the scale factor from the drawing to the room as a single number.

48

1 in : 4 ft is 1 in : 48 in

d. Find the room's real area in square metres.

About 17.86 m²

3.66 m by 4.88 m = 17.8608 m²; or 192 × 0.305²

PART D. Converting Length Across Systems; Currency *Lessons 6–7 · MA.7.AR.3.3 · 6 questions*

16. 1 in = 2.54 cm. How many centimetres is 14 in?

35.56 cm

17. 1 cm = 0.39 in. How many inches is 120 cm?

46.8 in (47.2 by ÷ 2.54)

fewer inches than centimetres

18. 1 km = 0.62 mi. How many miles is 30 km?

18.6 mi

19. 1 yd = 3 ft, 1 ft = 12 in, 1 in = 2.54 cm. How many centimetres is 6 yd? Estimate first.

About 540; exactly 548.64 cm

216 in × 2.54

20. Rate on 3 Nov: 1 US dollar = 0.92 euros.

a. How many euros is $150?

138 euros

× the rate

b. How many US dollars is 69 euros?

$75

÷ the rate

21. 1 in = 2.54 cm; 1 cm = 0.39 in. Lee converts 40 cm to inches by multiplying by 2.54 and gets 101.6 in. Name the error and fix it.

Lee used the ratio whose 1 is inches on centimetres. Inches are bigger, so there are fewer: 40 × 0.39 = 15.6 in (or 40 ÷ 2.54 = 15.7).

the error named and the estimate

PART E. Converting Mass, Area and Volume

Lessons 8–9 · MA.7.AR.3.3 · 5 questions

22. 1 lb = 0.454 kg. How many kilograms is 80 lb?

36.32 kg

(80 ÷ 2.204 = 36.30 by the other ratio)

23. 1 L = 0.264 gal. How many gallons is 40 L?

10.56 gal

24. 1 ft = 12 in. How many square inches is 8 ft^2?

1,152 in^2

8 × 144: the ratio squared

25. 1 c = 8 fl oz and 1 c = 0.24 L. How many litres is 48 fl oz?

1.44 L

6 c × 0.24

26. 1 m = 100 cm. A desk is 1.5 m^2 in area. How many square centimetres is that?

- ○ A. 300 cm^2
- ○ B. 150 cm^2
- ● **C. 15,000 cm^2**
- ○ D. 1,500 cm^2

C

PART F. Multi-Step Proportion Problems

Lessons 10–11 · MA.7.AR.4.5 · 4 questions

27. A car does 29 miles per gallon and its tank holds 12 gal.

a. How far does it go on a full tank?

348 mi

b. It starts an 800-mi trip with a full tank. How many tanks of gas does the trip take?

3 tanks (800 ÷ 348 ≈ 2.3)

a part of a tank is still a tank

28. A recipe for 3 servings uses 1¼ cups of rice.

a. How much rice for 12 servings?

5 cups

k = 4

b. With 10 cups on hand, how many servings can be made?

24 servings

8 recipes × 3

29. 1 km = 0.62 mi. A train does 110 km/h; a bus does 65 mi/h. Which is faster? Show both speeds in one unit.

The train: 110 km/h = 68.2 mi/h > 65 mi/h

same unit first

30. Paint is mixed 1 : 2 : 5. A batch uses 10 gal of the third colour. How many gallons is the whole batch?

16 gal

k = 2; 8 parts × 2